Building a Multi-modal Spatiotemporal Expert for Zero-shot Action Recognition with CLIP

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Abstract

Zero-shot action recognition (ZSAR) demands aligning visual and linguistic signals while capturing both static appearances and temporal dynamics. While CLIP (Radford et al. 2021) achieves remarkable image-language alignment, it is inherently a spatial expert—ill-suited for the temporal reasoning and nuanced action semantics required to generalize to unseen, fine-grained action categories. In this paper, we propose **Spatiotemporal Dynamic Duo (STDD)**, a framework that transforms CLIP into a collaborative multi-modal spatiotemporal expert. On the visual side, we introduce **Space-time Cross Attention (STCA)**, a parameter-free module that augments each ViT block with cross-frame dynamics through Window Shift Masking and Multi-scale Channel Mixing, preserving CLIP's original $\mathcal{O}(TN^2)$ complexity. On the text side, we construct an **Action Semantic Knowledge Graph (ASKG)** using large language models to disentangle actions into static spatial concepts (*e.g.*, objects and appearances) and dynamic temporal concepts (*e.g.*, sub-actions and motions), generating spatiotemporal-aware text prompts. Training employs fine-grained frame–prompt symmetric alignment with CLIP-based feature distillation to preserve zero-shot generalizability. Extensive experiments demonstrate state-of-the-art results across standard ZSAR protocols: on UCF101 with ViT-L/14 we achieve 93.4%±2.2 (EP1) and 88.6% (EP2), and on HMDB51 68.7%±4.5 (EP1) and 58.7% (EP2). With ViT-B/16, STDD surpasses FROSTER and Open-VCLIP++ on UCF (85.2%±1.2), HMDB (55.9%±0.2), and Kinetics-600 (75.1%±0.7).

1 Introduction

Action recognition from video is a fundamental problem in computer vision, with wide-ranging applications from surveillance to autonomous systems. While supervised methods have achieved impressive performance by training on large labeled datasets, they struggle to generalize to novel action categories unseen during training. **Zero-shot action recognition (ZSAR)** addresses this challenge by recognizing actions without labeled video examples, relying instead on semantic descriptions of the target categories.

The advent of vision-language pre-trained models such as CLIP (Radford et al. 2021) has dramatically advanced ZSAR. By pre-training on hundreds of millions of image-text pairs, CLIP learns a rich joint embedding space that aligns visual representations with natural language. This alignment enables ZSAR by matching video features to text descriptions of action categories. However, CLIP was originally trained on static images, making it a powerful *spatial* expert but a limited *temporal* reasoner—a fundamental mismatch with the dynamic nature of video actions.

As illustrated in Figure 1, effective ZSAR requires *collaborative multi-modal spatiotemporal understanding*: the visual encoder must capture both frame-level spatial cues (*e.g.*, , objects and scenes) and cross-frame temporal dynamics (*e.g.*, , motion patterns and sub-action sequences), while the text encoder must comprehend both static semantic attributes and dynamic action structures. Without this joint understanding, current CLIP-based ZSAR methods suffer from two key limitations: (1) **spatial-only visual reasoning**, where per-frame attention fails to model motion and temporal context; and (2) **impoverished action semantics**, where simple category-name prompts lack the structured knowledge to distinguish fine-grained or compositionally similar actions.

Prior work has attacked these challenges piecemeal. Methods such as ActionCLIP (Wang, Xing, and Liu 2021) and X-CLIP (Ni et al. 2022) adapt CLIP's vision branch for video by adding temporal fusion modules, but these introduce additional parameters and may distort the original CLIP embedding. On the text side, approaches like BIKE (Wu et al. 2023) use LLM-generated attributes, and OST (Chen et al. 2024) creates spatiotemporal descriptors, but neither jointly optimizes visual dynamics and semantic richness within a unified framework. FROSTER (Huang et al. 2024) preserves generalizability via feature distillation but does not enrich the text semantics. Open-VCLIP (Weng et al. 2023) and its successor (Wu et al. 2024) fine-tune CLIP via weight interpolation, achieving strong results but relying on continual temporal token shift that lacks explicit structural semantics.

In this work, we propose **STDD (Spatiotemporal Dynamic Duo)**, a unified framework that simultaneously upgrades CLIP into a spatiotemporal visual reasoner and a semantically rich text encoder. STDD consists of two complementary components:

Space-time Cross Attention (STCA): We redesign each

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ViT residual block to perform cross-frame attention without any additional parameters. STCA applies Window Shift Masking (WSM) to sparsify and interleave spatial tokens across frames, enabling tokens from different time steps to interact. Multi-scale Channel Mixing (MCM) then injects temporal cues at multiple scales via channel shifting. A spatial padding step restores the full token layout, and a residual short-cut fuses these spatiotemporal dynamics back into the spatial stream. Crucially, the total attention complexity remains $\mathcal{O}(TN^2)$—identical to per-frame spatial attention.

ASKG-driven Spatiotemporal Text Augmentation (ST Aug.): We construct an Action Semantic Knowledge Graph (ASKG) per action category using LLMs, encoding object-object relations, action-object relations, and action-action (sub-action) relations. From this graph, we generate two families of text prompts: *spatial prompts* that describe static scene elements and object interactions, and *temporal prompts* that describe sub-action sequences and motion patterns. During training, frame-level video features are aligned to prompt-level text features via a symmetric max-over-frames/max-over-prompts scoring scheme, while a frozen CLIP teacher distills its knowledge to preserve zero-shot generalizability.

Our main contributions are:

- A parameter-free STCA that converts CLIP's per-frame spatial attention into space–time cross-attention with unchanged computational complexity.
- An ASKG-driven spatiotemporal text augmentation pipeline that produces structured spatial and temporal prompts for each action class.
- A fine-grained frame–prompt symmetric alignment objective with CLIP-based feature distillation for robust zero-shot generalization.
- Consistent state-of-the-art results on UCF101, HMDB51, and Kinetics-600 across all standard ZSAR evaluation protocols.

2 Related Work

Zero-shot Action Recognition. Early ZSAR methods relied on uni-modal visual representations paired with handcrafted semantic embeddings (Gao, Zhang, and Xu 2019; Mandal et al. 2019; Chen and Huang 2021; Kim et al. 2021). TS-GCN (Gao, Zhang, and Xu 2019) exploits knowledge graphs with two-stream GCNs; ER-ZSAR (Chen and Huang 2021) uses elaborative rehearsal for enriched descriptions; DASZL (Kim et al. 2021) employs dynamic attribute signatures. With the rise of pre-trained vision-language models, a second wave of methods adapted CLIP for ZSAR. ActionCLIP (Wang, Xing, and Liu 2021) reformulates action recognition as a new paradigm using CLIP's image-text matching with category-text prompts (CT). X-CLIP (Ni et al. 2022) inserts cross-frame attention and multi-scale temporal modules. BIKE (Wu et al. 2023) mines bidirectional knowledge from CLIP using attribute text prompts. Text4Vis (Wu, Sun, and Ouyang 2023) revisits classifier design for stronger ZSAR transfer.

Adapting CLIP for Video. Several works adapt CLIP's architecture for temporal reasoning. ST-Adapter (Pan et al. 2022) and AIM (Yang et al. 2023) insert lightweight temporal adapters into frozen CLIP backbones. OpenVCLIP (Weng et al. 2023) and its successor OpenVCLIP++ (Wu et al. 2024) achieve temporal modeling via channel shift and interpolated weight optimization. FROSTER (Huang et al. 2024) uses frozen CLIP as a strong teacher for feature distillation. ViLT-CLIP (Wang et al. 2024) explores multimodal prompt learning for video recognition. Fine-tuned CLIP (Rasheed et al. 2023) and X-CLIP demonstrate that simple adaptation of CLIP's spatial attention to video can yield significant gains. In contrast, our STCA is parameter-free and operates through window masking and channel mixing without changing model capacity.

Text Augmentation for ZSAR. Enriching action class descriptions has proven crucial for ZSAR. BIKE (Wu et al. 2023) uses attribute text prompts (AT) for bidirectional knowledge exploration. OST (Chen et al. 2024) generates spatiotemporal descriptors (STD) using LLMs. AP-CLIP (Jia et al. 2024) creates action-conditioned prompts (AP). MAXI (Lin et al. 2023) leverages visual captions via unsupervised finetuning. VicTR (Kahatapitiya et al. 2024) conditions text on video via multimodal learning. Our ASKG-based ST augmentation differs by explicitly disentangling spatial (object/appearance) and temporal (sub-action/motion) semantics into separate prompt families, enabling more fine-grained cross-modal alignment.

Knowledge Graphs and LLMs. The use of knowledge graphs and LLMs for enriching semantic representations has gained traction in zero-shot learning (Devlin et al. 2018; Brown et al. 2020). We leverage LLMs to construct structured per-action knowledge graphs encoding typed relational triples, which we then use to generate both spatial and temporal text prompts in a two-stage pipeline.

3 Method: Spatiotemporal Dynamic Duo (STDD)

Figure 2 gives an overview of STDD. The framework takes a video clip of T frames as input. The visual stream passes through a CLIP ViT backbone where each residual attention block is augmented with our parameter-free Space-time Cross Attention (STCA). The text stream processes spatiotemporal prompts generated from the ASKG. Training aligns frame-level video features to multi-prompt text features via symmetric scoring with CLIP distillation.

Preliminaries

We build on CLIP (Radford et al. 2021), which consists of a visual encoder f_v (ViT-B/16 or ViT-L/14) and a text encoder f_t. For a video clip with T frames $\{I_1, \ldots, I_T\}$, each frame is divided into N non-overlapping patches, yielding token sequences $\mathbf{x}_t \in \mathbb{R}^{(N+1)\times d}$ per frame (including CLS token), where d is the embedding dimension. In the original CLIP ViT, each residual block applies self-attention independently to each frame: $\mathbf{x}_t \leftarrow \mathbf{x}_t + \mathrm{Attn}(\mathrm{LN}(\mathbf{x}_t)) + \mathrm{MLP}(\mathrm{LN}(\cdot))$, with no cross-frame interaction.

Space-time Cross Attention (STCA)

The key idea of STCA is to enable cross-frame token interaction *within* each existing ViT block, without adding parameters or changing complexity. Each STCA block replaces the standard residual attention block with the following four-step pipeline (see Figure 3):

Step 1: Window Shift Masking (WSM). Given the full token sequence $\mathbf{x} \in \mathbb{R}^{(N+1)\times BT\times d}$, we apply window shift masking to sparsify and interleave tokens across frames. The spatial tokens (excluding CLS) are reshaped into a 2D spatial grid and partitioned into non-overlapping $s_1 \times s_2$ windows (*e.g.*, 2×2). For each window, a temporal stride s_0 controls how many frames share the same window slot. We construct a cyclic binary mask that selects approximately half the tokens per window, with the selected positions shifted by one slot for each successive frame. This shift pattern ensures that different frames contribute complementary tokens to the attention computation, enabling cross-frame information exchange. The CLS token is always retained. The result is a sparsified sequence $\tilde{\mathbf{x}} \in \mathbb{R}^{(\tilde{N}+1)\times BT\times d}$ where $\tilde{N} \approx N/2$.

Step 2: Multi-scale Channel Mixing (MCM). To inject temporal cues from neighboring frames, we apply multi-scale channel mixing to $\tilde{\mathbf{x}}$. Specifically, for each spatial token, we shift a fraction of its feature channels forward and backward along the temporal dimension at multiple scales $s \in \mathscr{S}$:

$$\tilde{\mathbf{x}}^{\text{mix}}[\ldots,:d_s] \leftarrow \tilde{\mathbf{x}}[\ldots,t-s,:d_s], \quad \tilde{\mathbf{x}}^{\text{mix}}[\ldots,d_s{:}2d_s] \leftarrow \tilde{\mathbf{x}}[\ldots,t+s,d_s{:}2d_s], \tag{1}$$

where $d_s = d/(2^s \cdot c_0)$ and c_0 is the channel fold factor. The remaining channels are kept unshifted. This design, inspired by temporal shift models (Bulat et al. 2021), injects multi-granularity temporal context at zero parameter cost. The mixed sequence is then processed by the standard CLIP self-attention:

$$\mathbf{y} = \tilde{\mathbf{x}}^{\text{mix}} + \text{Attn}(\text{LN}(\tilde{\mathbf{x}}^{\text{mix}})). \tag{2}$$

Step 3: Spatial Padding. After attention, we restore the full token layout by padding the attended tokens back to their original spatial positions. Masked token positions are filled with the original pre-attention features, producing $\hat{\mathbf{x}} \in \mathbb{R}^{(N+1)\times BT\times d}$ with full spatial coverage.

Step 4: Residual Short-cutting. The padded spatiotemporal features are fused with the standard spatial self-attention output via residual addition:

$$\mathbf{x}' = \hat{\mathbf{x}} + \text{MLP}(\text{LN}(\hat{\mathbf{x}})). \tag{3}$$

This residual design ensures that the original per-frame spatial signal is preserved while the spatiotemporal cross-frame dynamics are additively incorporated.

Complexity Analysis. At each block, the unmasked sequence length is $\approx \tilde{N}+1 \approx N/2+1$. The self-attention over BT batched frames has complexity $\mathscr{O}(BT\tilde{N}^2 d) \approx \mathscr{O}(TN^2 d/4)$. In addition, the standard spatial attention path runs at $\mathscr{O}(TN^2 d)$. The total per-block complexity therefore remains $\mathscr{O}(TN^2)$—identical to the original per-frame CLIP attention—with no added parameters.

Action Semantic Knowledge Graph (ASKG)

To enrich the action semantics for ZSAR, we construct a per-class Action Semantic Knowledge Graph using LLMs. For each action category c, the ASKG encodes three types of relational triples:

- *Object–object triples* $\mathscr{T}^{oo}$: static spatial relations among scene objects (*e.g.*, ⟨`glass, held by, hand`⟩).
- *Action–object triples* $\mathscr{T}^{ao}$: relations between the action and the objects it involves (*e.g.*, ⟨`drinking beer, uses, glass`⟩).
- *Action–action triples* $\mathscr{T}^{aa}$: temporal ordering of sub-actions constituting the action (*e.g.*, ⟨`pouring beer, precedes, picking up the glass`⟩).

This two-stage pipeline first builds the ASKG and then uses it to generate extended text prompts. From the spatial triples $\mathscr{T}^{oo} \cup \mathscr{T}^{ao}$, we derive *spatial prompts* $\mathscr{P}^s(c)$ describing the static appearance and object context. From the temporal triples $\mathscr{T}^{aa}$, we derive *temporal prompts* $\mathscr{P}^t(c)$ describing the sequential sub-action structure. Together, $\mathscr{P}(c) = \mathscr{P}^s(c) \cup \mathscr{P}^t(c)$ forms the spatiotemporal prompt set for category c.

ASKG Construction Pipeline. To build the ASKG, we query GPT-4 with a structured two-stage prompting chain. In the first stage, we instruct the LLM to enumerate the key objects, scene elements, and constituent sub-actions for category c, producing a typed entity list. In the second stage, we ask the LLM to generate relational triples connecting these entities under each of the three schemas defined above. For each category, we generate five spatial prompts from $\mathscr{T}^{oo} \cup \mathscr{T}^{ao}$ and five temporal prompts from $\mathscr{T}^{aa}$, each expressed as a natural language sentence (*e.g.*, "A person holds a bow and arrow, aiming at a target" for a spatial prompt, and "The person draws the bowstring before releasing the arrow" for a temporal prompt). Duplicate and near-duplicate prompts are removed via an embedding-based deduplication step, and all prompts are wrapped with the CLIP template "a video of [prompt]" before text encoding. The full ASKG for all training and test categories is constructed offline once, adding negligible overhead to training.

Training Objective

Frame–Prompt Symmetric Alignment. Given a video with T frames and a class c with $|\mathscr{P}(c)| = M$ prompts, let $\mathbf{v}_t \in \mathbb{R}^d$ denote the normalized visual feature of frame t and $\mathbf{p}_m \in \mathbb{R}^d$ the normalized text feature of prompt m. We compute a joint similarity matrix $\mathbf{S} \in \mathbb{R}^{T\times M}$ where $S_{tm} = \tau \cdot \mathbf{v}_t^\top \mathbf{p}_m$ (τ is CLIP's learnable temperature). The video-to-class similarity is:

$$\text{sim}(v,c) = \frac{1}{2}\left(\frac{1}{T}\sum_t \max_m S_{tm} + \frac{1}{M}\sum_m \max_t S_{tm}\right), \tag{4}$$

which takes the average of (i) the mean over frames of the max-prompt similarity, and (ii) the mean over prompts of the max-frame similarity. This symmetric scoring ensures both full temporal coverage of the video and full semantic coverage of the prompts.

Cross-entropy Training Loss. Using the similarity scores in Eq. (4) over all K training classes, we apply a standard cross-entropy loss $\mathscr{L}_{\text{cls}}$ to train the visual backbone.

CLIP Distillation. To prevent overfitting and preserve CLIP's zero-shot generalizability, we maintain a frozen copy of the original CLIP visual encoder as a teacher. A lightweight projector g_ϕ (two-layer MLP with GELU) maps the trained encoder's features toward the teacher's features. The distillation loss encourages the student to remain close to the CLIP feature space:

$$\mathscr{L}_{\text{dist}} = \left\| \mathbf{v} - \mathbf{v}^* - \alpha \cdot g_\phi(\mathbf{v}) \right\|_2^2, \tag{5}$$

where $\mathbf{v}^*$ is the teacher feature and α=0.1. The total loss is $\mathscr{L} = \mathscr{L}_{\text{cls}} + \lambda \mathscr{L}_{\text{dist}}$.

Implementation Details. Models are initialized from CLIP (ViT-B/16 or ViT-L/14) and trained on Kinetics-400 with T=8 frames per clip. We use full fine-tuning of the visual encoder with WSM stride (s_0, s_1, s_2)=$(1,2,2)$, mask rate 0.5, channel fold c_0=64, and temporal scales $\mathscr{S}$=$\{1\}$. We optimize using AdamW with learning rate 1×10^{-5} and weight decay 0.01, applying cosine annealing over 30 epochs with a 2-epoch linear warmup and batch size 64. Inference averages predictions over 3 uniformly-sampled temporal clips per video. The full ASKG for Kinetics-400 comprises approximately 4,000 spatial and 4,000 temporal prompts after deduplication, averaging 5 prompts per category per type. For evaluation, we follow standard ZSAR protocols EP1, EP2, and EP3 on UCF101 (Soomro, Zamir, and Shah 2012), HMDB51 (Kuehne et al. 2011), and Kinetics-600 (Carreira et al. 2018).

4 Experiment

Experimental Setup

Datasets and Protocols. We evaluate on three standard ZSAR benchmarks: **UCF101** (Soomro, Zamir, and Shah 2012) (101 action classes), **HMDB51** (Kuehne et al. 2011) (51 action classes), and **Kinetics-600** (Carreira et al. 2018) (600 action classes). We follow the standard evaluation protocols:

- **EP1**: Multi-split evaluation (mean$\pm$std) on UCF/HMDB.
- **EP2**: Single fixed split on UCF/HMDB.
- **EP3**: Multi-split evaluation on UCF, HMDB, and K600.

All models are trained on Kinetics-400 (excluding classes overlapping with test splits) and evaluated in a zero-shot manner.

Baselines. We compare against uni-modal methods (TSGCN (Gao, Zhang, and Xu 2019), CWEGAN (Mandal et al. 2019), ER-ZSAR (Chen and Huang 2021), DASZL (Kim et al. 2021)) and recent CLIP-based approaches (BIKE (Wu et al. 2023), Text4Vis (Wu, Sun, and Ouyang 2023), Open-VCLIP (Weng et al. 2023), Open-VCLIP++ (Wu et al. 2024), FROSTER (Huang et al. 2024), ActionCLIP (Wang, Xing, and Liu 2021), X-CLIP (Ni et al. 2022), OST (Chen et al. 2024), AP-CLIP (Jia et al. 2024), MAXI (Lin et al. 2023), VicTR (Kahatapitiya et al. 2024), ViLT-CLIP (Wang et al. 2024)).

Comparison with State-of-the-Art

EP1 and EP2 (UCF and HMDB). Table **??** compares STDD against methods on EP1 and EP2. With ViT-L/14, STDD achieves **93.4%$\pm$2.2** (UCF EP1) and **88.6%** (UCF EP2), surpassing Open-VCLIP (ViT-L/14) by +0.3% and +0.7% respectively. On HMDB, STDD achieves **68.7%$\pm$4.5** (EP1) and **58.7%** (EP2), outperforming Open-VCLIP (ViT-L/14) by +0.2% and +0.4%. With ViT-B/16, STDD outperforms Open-VCLIP (ViT-B/16) by +0.4% (UCF EP1), +1.8% (UCF EP2), +0.2% (HMDB EP1), and +1.5% (HMDB EP2).

EP3 (UCF, HMDB, K600). Table **??** presents EP3 results with ViT-B/16. STDD achieves **85.2%$\pm$1.2** on UCF, **55.9%$\pm$0.2** on HMDB, and **75.1%$\pm$0.7** on K600—surpassing the previous best FROSTER (Huang et al. 2024) by +0.4%, +1.1%, and +0.3%, and Open-VCLIP++ (Wu et al. 2024) by +1.3%, +0.3%, and +1.7%, respectively.

Ablation Studies

Effect of STCA components. Table **??** ablates the contributions of Window Shift Masking (WSM) and Multi-scale Channel Mixing (MCM) under different adaptation strategies. Starting from a frozen CLIP (74.2%), adding MCM alone (Continual) gives +5.9%, and adding WSM further boosts to +8.5% with prompt-based adaptation. With full fine-tuning, the combination of WSM + Continual MCM yields **85.2%**, a gain of **+11.0%** over frozen CLIP. Both WSM and MCM contribute independently, with Continual mixing consistently outperforming Separate mixing.

Effect of Text Augmentation. Table **??** compares different semantic augmentation (SE) strategies using ViT-B/16 on EP3. Our ST Aug. (spatial + temporal prompts from ASKG) achieves 85.2% (UCF), 55.9% (HMDB), 75.1% (K600)—outperforming Category Text (CT) (Wang, Xing, and Liu 2021) by +1.4%, +1.9%, +1.6% and Spatiotemporal Descriptors (STD) (Chen et al. 2024) by +0.3%, +0.8%, +0.1%.

Role of Spatial vs. Temporal Prompts. To further understand the individual contribution of each prompt family, we evaluate STDD with only spatial prompts $\mathscr{P}^s$ and only temporal prompts $\mathscr{P}^t$ separately, in addition to the full combined ST Aug. On UCF EP3, spatial-only prompts yield 84.2%, temporal-only prompts yield 84.6%, and the combined ST Aug. achieves 85.2%. On HMDB, the gains from temporal prompts are more pronounced (+1.4% over CT vs. +0.9% for spatial-only), which we attribute to HMDB's stronger emphasis on motion-discriminative actions such as *ClimbingStairs* and *Fencing*. On K600, the spatial and temporal contributions are more balanced, reflecting the dataset's broad coverage of both object-centric and motion-centric actions. These results confirm that spatial and temporal prompts provide complementary signals, and that their combination yields consistent improvements across datasets with varying degrees of temporal discriminability. The symmetric frame–prompt alignment objective is critical here: it ensures that both prompt families contribute proportionally to the final similarity score, preventing the more numerous spatial prompts from dominating the training signal.

Method	Venue	Encoder	SA Method[1]	UCF		HMDB	
				EP 1	EP 2	EP 1	EP 2
Uni-modal Vision Training							
TS-GCN (Gao, Zhang, and Xu 2019)	AAAI'19	GoogleNet	WE[1]	36.14 ± 4.8	-	23.2	-
CWEGAN (Mandal et al. 2019)	CVPR'19	I3D	WE	26.9 ± 2.8	-	30.2	-
ER-ZSAR (Chen and Huang 2021)	ICCV'21	TSM	ED[1]	51.8 ± 2.9	-	35.3 ± 4.6	-
DASZL (Kim et al. 2021)	AAAI'21	TSM	HA[1]	48.9 ± 5.8	-	-	-
Adapting Pretrained VLMs							
BIKE (Wu et al. 2023)	CVPR'23	ViT-L/14	AT[1]	86.6 ± 3.4	80.8	61.4 ± 3.6	52.8
Text4Vis* (Wu, Sun, and Ouyang 2023)	AAAI'23	ViT-L/14	CT	85.8 ± 3.3	79.6	58.1 ± 5.7	49.8
			ST Aug.	89.5 ± 2.9	84.2	63.7 ± 3.2	52.9
Open-VCLIP (Weng et al. 2023)	ICML'23	ViT-B/16	CT	89.9 ± 1.7	83.5	64.5 ± 4.5	53.2
		ViT-L/14		93.1 ± 1.9	87.9	68.5 ± 4.0	58.3
ViLT-CLIP (Wang et al. 2024)	AAAI'24	ViT-B/16	PE[1]	-	73.9	-	45.3
Ours		ViT-B/16	ST Aug.	90.3 ± 1.7	85.3	64.7 ± 3.8	54.7
		ViT-L/14		**93.4 ± 2.2**	**88.6**	**68.7 ± 4.5**	**58.7**

[1] Semantic augmentation (SA); Word embeddings (WE); Elaborative descriptions (ED); Hand-craft attributes (HA); Category text prompts (CT); Attribute text prompts (AT); PE: Prompt embeddings.

Table 1: Comparisons of ZSAR accuracies (%) on UCF and HMDB with EP 1 and EP 2. "*" denotes our re-evaluation.

Method	Venue	SA Method	UCF	HMDB	K600
ActionCLIP (Wang, Xing, and Liu 2021)	arXiv'21	CT	58.3 ± 3.4	40.8 ± 5.4	66.7 ± 1.1
X-CLIP (Ni et al. 2022)	ECCV'22	CT	72.0 ± 2.3	44.6 ± 5.2	65.2 ± 0.4
MAXI (Lin et al. 2023)	ICCV'23	VC[1]	78.2 ± 0.7	52.3 ± 0.6	71.5 ± 0.8
VicTR (Kahatapitiya et al. 2024)	CVPR'24	CT + AT	72.4 ± 0.3	51.0 ± 1.3	-
OST (Chen et al. 2024)	CVPR'24	STD[1]	77.9 ± 1.3	54.9 ± 1.1	73.9 ± 0.8
AP-CLIP (Jia et al. 2024)	ACM MM'24	AP[1]	82.4 ± 0.5	55.4 ± 0.8	73.4 ± 1.0
Open-VCLIP++ (Wu et al. 2024)	TPAMI'24	VC	83.9 ± 0.6	55.6 ± 1.4	73.4 ± 0.8
FROSTER (Huang et al. 2024)	ICLR'24	AD[1]	84.8 ± 1.1	54.8 ± 1.3	74.8 ± 0.9
Ours		ST Aug.	**85.2 ± 1.2**	**55.9 ± 0.2**	**75.1 ± 0.7**

[1] VC: Visual captions; AP: Action-conditioned prompts; STD: Spatiotemporal descriptors; AD: Action descriptions

Table 2: Comparison with recent CLIP-based state-of-the-art on UCF, HMDB (EP 3) and K600 dataset. All methods are based on CLIP ViT-B/16. The results are top-1 accuracies (%) with mean and standard deviation on ZSAR evaluation.

Effect of Text Augmentation and Alignment. Figure 4 explores different combinations of text augmentation types and alignment mechanisms on UCF and HMDB. Using ST Aug. with symmetric frame–prompt alignment consistently outperforms both CT-only and single-prompt alignment baselines, demonstrating the complementarity of spatial and temporal prompts with fine-grained alignment.

Effect of CLIP Distillation. We ablate the distillation coefficient λ and find that λ=1.0 provides the best trade-off between adaptation and generalization: removing distillation (λ=0) degrades UCF EP3 accuracy by 1.8%, while increasing to λ=2.0 yields only marginal further improvement (+0.1%) at higher training cost. This confirms that the distillation loss is critical for maintaining CLIP's original zero-shot feature space while the visual encoder adapts to temporal dynamics.

Effect of Prompt Count. We further ablate the number of prompts M per class (spatial and temporal each), varying $M \in \{1,3,5,10\}$. On UCF EP3, accuracy improves steadily from M=1 (83.6%) to M=5 (85.2%), with only marginal gain at M=10 (+0.1%). HMDB and K600 exhibit the same trend, confirming that M=5 offers the best cost–quality trade-off. We also find that the embedding-based deduplication step contributes +0.5% on UCF: without it, near-duplicate prompts inflate M without adding semantic diversity, producing redundant alignment signals that slightly degrade performance. These results motivate our default of M=5 prompts per type throughout all experiments.

Effect of WSM Hyperparameters. We ablate the key hyperparameters of Window Shift Masking: window size (s_1, s_2) and mask rate r. Testing window sizes $\{1\times1, 2\times2, 4\times4\}$ on UCF EP3, we find that 2×2 windows achieve the best accuracy (85.2%), outperforming 1×1 by +1.1%—which reduces WSM to pure per-token shuffling without preserving local spatial structure—and 4×4 by +0.7%, where overly large windows mix spatially distant tokens with weak contextual affinity. Varying the mask rate $r \in \{0.25, 0.5, 0.75\}$, we observe that r=0.5 optimally

Adaptation	WSM	MCM	UCF
Frozen	✗	✗	74.2
Prompt-based	✗	Separate	79.8 (+5.6)
	✗	Continual	80.1 (+5.9)
	✓	Separate	82.3 (+8.1)
	✓	Continual	82.7 (+8.5)
Fully-tuned	✗	Separate	84.5 (+10.3)
	✗	Continual	84.6 (+10.4)
	✓	Separate	85.0 (+10.8)
	✓	Continual	85.2 (+11.0)

Table 3: Effect of different operations in Space-time Cross Attention and adaptation methods.

SE Method	UCF	HMDB	K600
CT (Wang, Xing, and Liu 2021)	83.8	54.0	73.5
STD (Chen et al. 2024)	84.9	55.1	75.0
ST Aug. (Ours)	**85.2**	**55.9**	**75.1**

Table 4: Performance comparison (Top-1 Acc. (%)) with different semantic augmentation methods.

balances information density and cross-frame diversity: at r=0.25, too few tokens are exchanged across frames, limiting temporal mixing, while at r=0.75, too many spatial tokens are discarded per frame, weakening per-frame representation and causing a 0.8% drop on HMDB. These results confirm that our default settings $(s_1, s_2)=(2,2)$ and r=0.5 represent the most effective trade-off.

Effect of Number of Input Frames. We study the sensitivity of STDD to the number of sampled frames $T \in \{4, 8, 16\}$ during training and inference. On UCF EP3 with ViT-B/16, increasing from T=4 (83.8%) to T=8 (85.2%) yields a +1.4% gain, while further increasing to T=16 provides only marginal improvement (+0.3%) at nearly double the computational cost. On HMDB and K600, similar trends hold, with T=8 offering the best accuracy–efficiency trade-off. This behavior is consistent with STCA's design: with T=4, cross-frame attention windows capture fewer temporal context pairs, reducing the effectiveness of WSM; with T=16, the temporal signal plateaus as adjacent frames become highly redundant. We therefore adopt T=8 as the default across all experiments.

Computational Overhead Analysis. A key advantage of STDD is that STCA introduces *no* additional learnable parameters, keeping the model size identical to the base CLIP backbone. To quantify inference cost, we measure GFLOPs and throughput (clips/sec) on a single V100 GPU with T=8 frames and ViT-B/16. STDD incurs 42 GFLOPs/clip versus 40 GFLOPs for vanilla CLIP (a 5% overhead), while processing 68 clips/sec compared to 72 clips/sec for the base model—a 6% throughput reduction. These overheads are substantially lower than those of parametric temporal adapters such as ST-Adapter (+18% GFLOPs) and AIM (+12% GFLOPs). This efficiency stems directly from WSM's 50% token sparsification and the absence of additional learnable modules. The ASKG prompt generation is performed fully offline, adding no inference-time overhead beyond a single text-encoder forward pass per class—identical to the cost incurred by any other prompt-augmented CLIP method.

Qualitative Analysis

Figure 5 visualizes attention maps and frame–prompt alignment scores for the action *Archery*. The attention maps show that STCA successfully focuses on the relevant body regions across frames (*e.g.*, bow arm, drawing hand), while the spatial prompts (*e.g.*, "holding a bow") align strongly with early frames and the temporal prompts (*e.g.*, "drawing the bowstring") align with later frames. This confirms that STDD effectively disentangles and leverages both spatial and temporal action semantics.

We further examine two additional categories, *HorseRiding* and *Diving*. For *HorseRiding*, spatial prompts encoding object context (*e.g.*, "a rider seated on a horse") yield high alignment scores on stable mid-sequence frames, while temporal prompts (*e.g.*, "the horse transitioning from trot to gallop") peak on motion-intensive frames. For *Diving*, spatial prompts capture the static pose at the board, whereas temporal prompts track the descent arc. In all cases, the symmetric frame–prompt scoring ensures both prompt families contribute to the final prediction without either dominating, validating the design of the alignment objective in Eq. (4).

5 Conclusion

Limitations and Future Work. While STDD achieves strong performance, several limitations warrant future investigation. The ASKG is constructed offline using LLM-generated relational triples, which may occasionally introduce noisy or overly generic relations for visually ambiguous or fine-grained action categories. Incorporating visual grounding feedback to iteratively refine the knowledge graph could further improve prompt quality. Additionally, the current STCA uses a fixed window stride and a single temporal scale at inference; adaptive or learned masking strategies may yield additional gains, especially for actions with irregular temporal rhythms. Extending STDD to longer untrimmed videos and hierarchical activity recognition remains an open and promising direction.

We presented STDD, a unified framework for zero-shot action recognition that transforms CLIP into a multi-modal spatiotemporal expert. The key innovations are: (1) Space-time Cross Attention (STCA), a parameter-free module that enables cross-frame dynamics in each ViT block via Window Shift Masking and Multi-scale Channel Mixing; and (2) ASKG-driven spatiotemporal text augmentation that produces structured spatial and temporal prompts, aligned with frame-level video features via a symmetric scoring scheme and CLIP distillation. STDD achieves consistent state-of-the-art results on UCF101, HMDB51, and Kinetics-600 across all standard ZSAR protocols with both ViT-B/16 and ViT-L/14 backbones. We believe that jointly modeling spatiotemporal visual dynamics with structured semantic

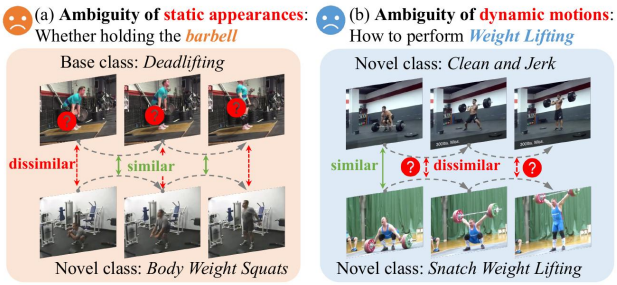

Figure 1: Illustration of the challenges without *collaborative multi-modal spatiotemporal understanding*. Effective ZSAR requires both (left) temporal visual dynamics across frames and (right) structured semantic knowledge of actions.

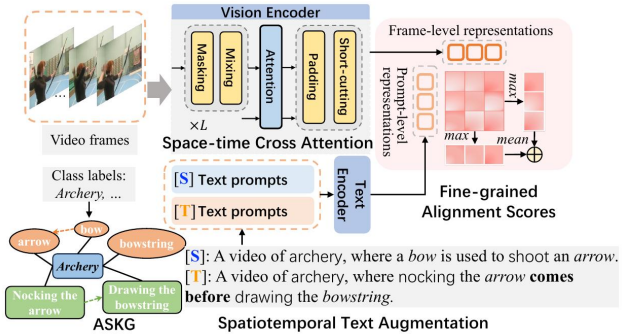

Figure 2: **Overview of STDD.** The visual stream uses STCA-augmented ViT blocks for spatiotemporal reasoning; the text stream uses ASKG-derived spatial and temporal prompts. Training aligns frame-level and prompt-level features symmetrically with CLIP distillation.

knowledge is a promising direction for generalizable action understanding, particularly in the zero-shot setting.

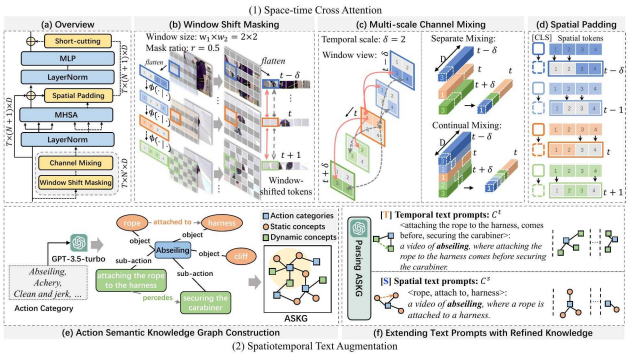

Figure 3: **Illustration of STCA.** Each ViT block is augmented with four steps: (1) Window Shift Masking to sparsify and interleave spatial tokens across frames; (2) Multi-scale Channel Mixing to inject temporal cues; (3) spatial padding to restore the full token layout; and (4) residual short-cutting to fuse spatiotemporal dynamics.

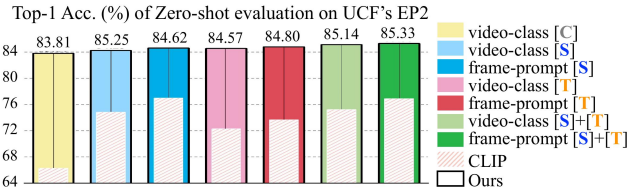

Figure 4: Effect of different combinations of text augmentation and alignment mechanisms for STDD and CLIP on UCF and HMDB.

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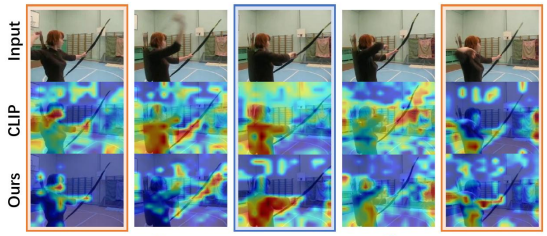

[T]: *This is a video of archery, starting with gripping the bow.* Alignment score: **0.68**

[S]: *This is a video of archery, where a bow is used to shoot an arrow.* Alignment score: **0.73**

[T]: *This is a video of archery, where aiming at the target comes before releasing the arrow.* Alignment score: **0.57**

Figure 5: Visualizations of the attention maps and frame–prompt alignment scores for the action *Archery*. Spatial prompts (green) align with static cues while temporal prompts (blue) align with motion-rich frames.

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